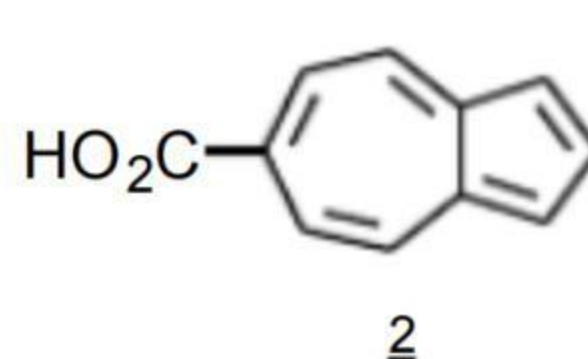
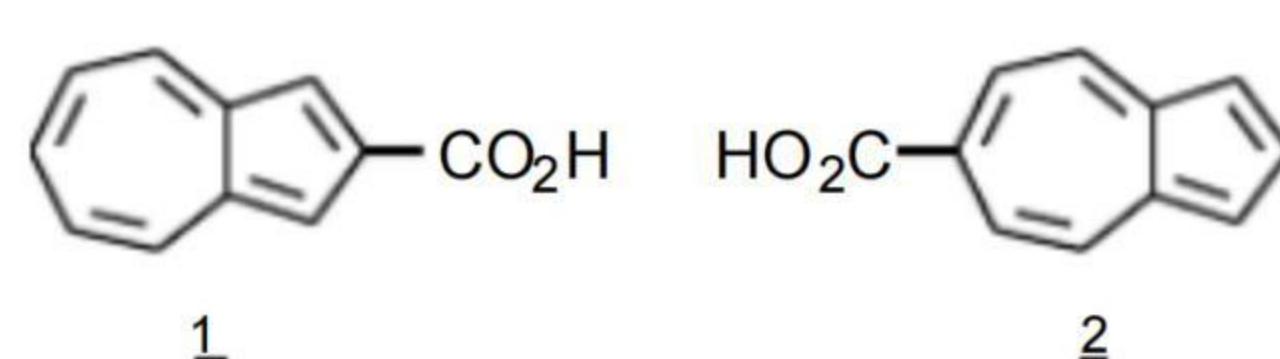
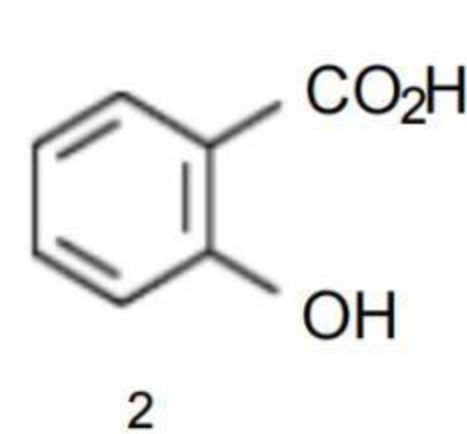
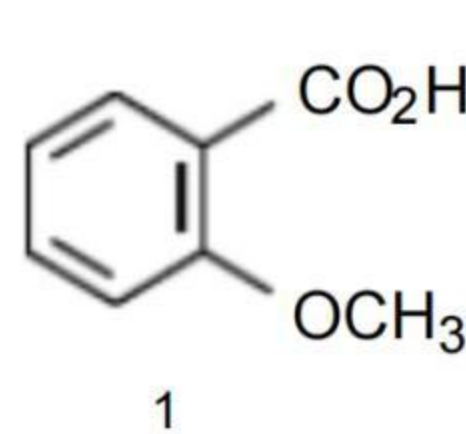


华东理工大学 2017-2018学年第二学期 《有机化学》 期终试卷 (A)

班级_____ 姓名_____ 学号_____ 成绩_____

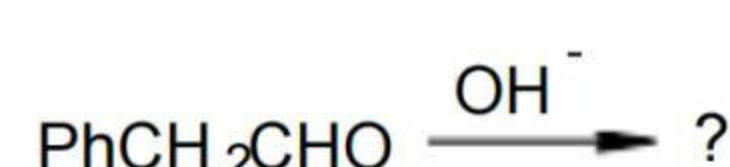
(用英文或中文回答下列六大题)

1. Order the following compounds in each set with respect to increasing acidity and explain why? (10 points)

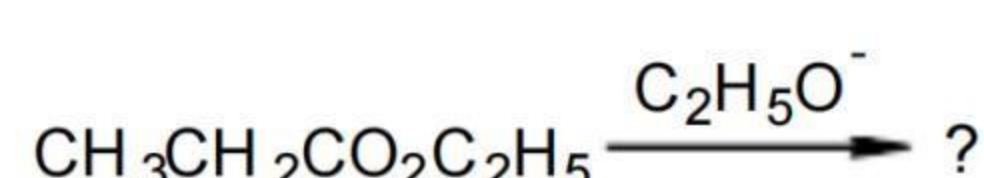


2. Predict the reagent or product of the following reactions:(20 points):

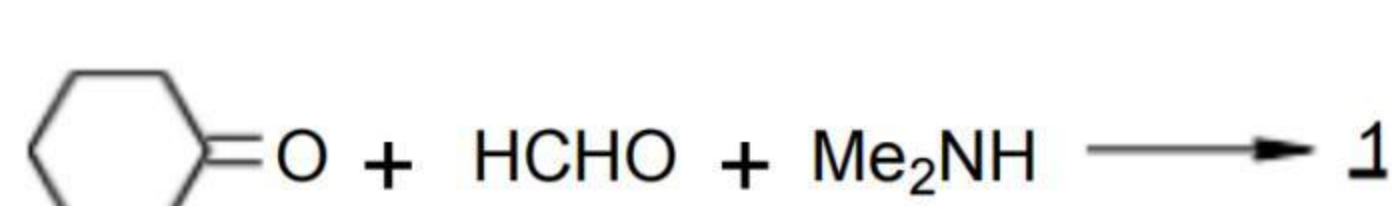
1)



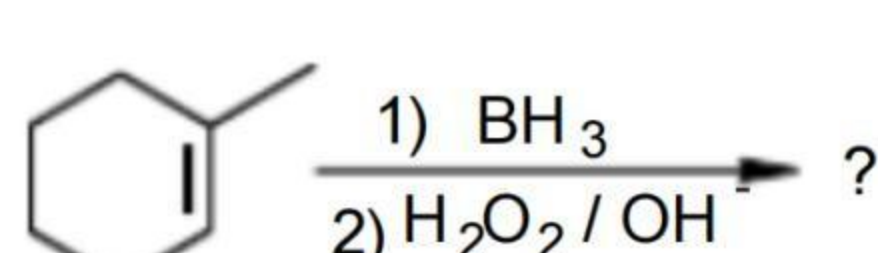
2)



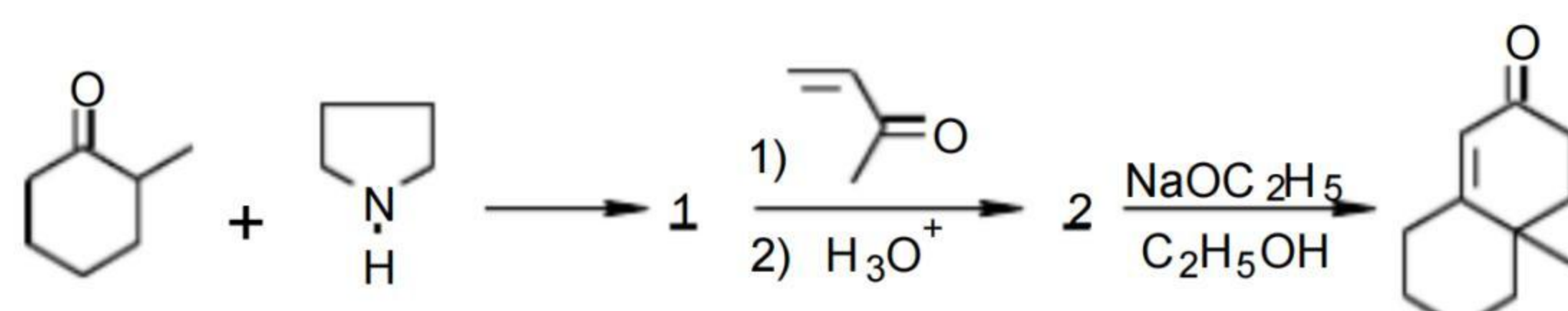
3)



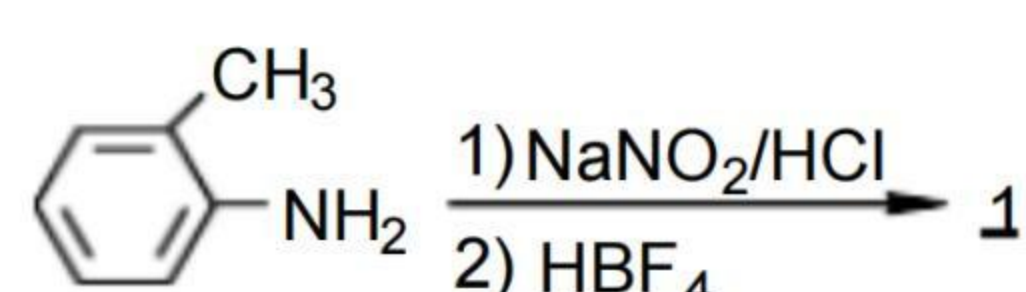
4)



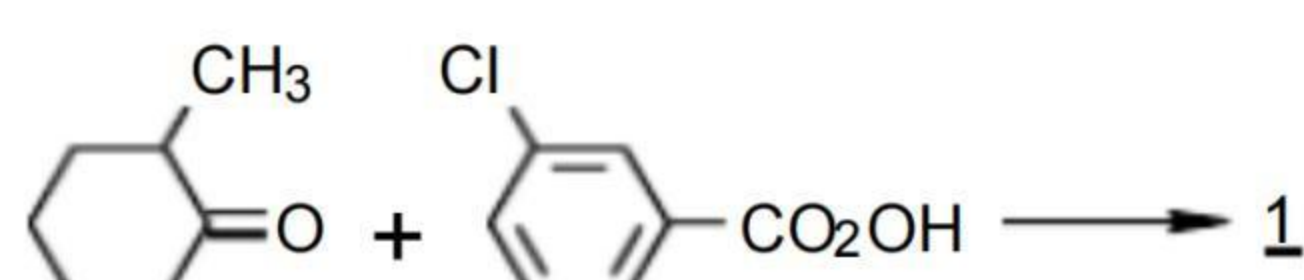
5)



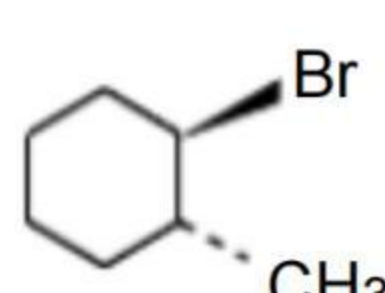
6)



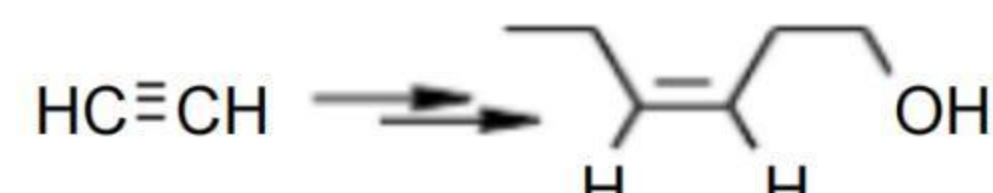
7)



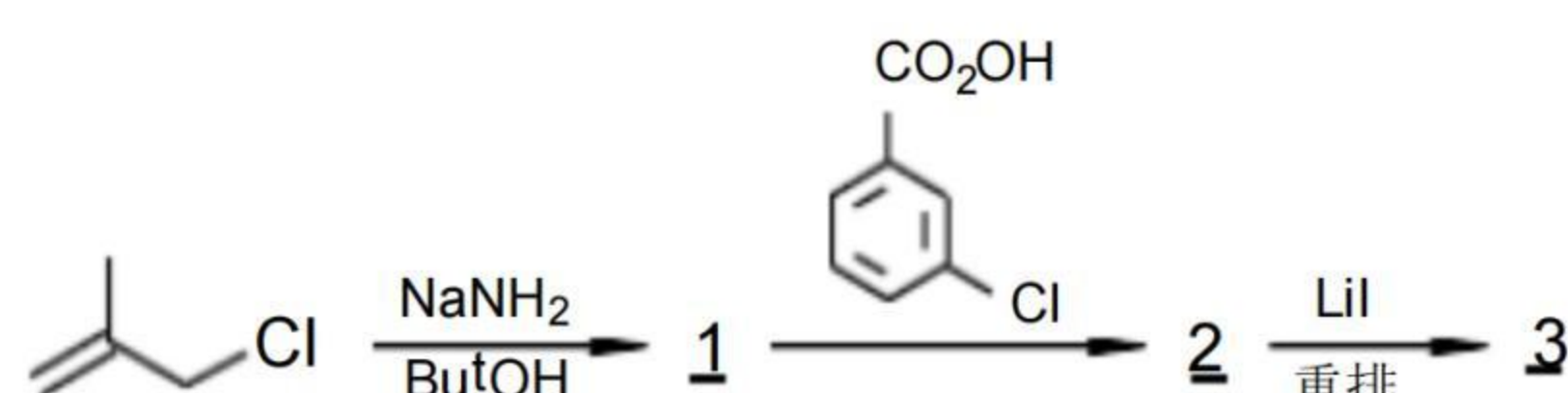
3. For following compound, Give :1) Chinese name (the R/S designation of each chiral center should be concerned) of IUPAC 2) equilibrating chair conformation and indicate which one is the most stable 3) the product from the reaction with strong base by E2 process.(12 points)



4. 1) Give three methods for converting a ketone group to methylene group
2) How would you prepare the following two compounds from the starting material given? (33 points)



5. Three compounds 1 (C₄H₆), 2 (C₄H₆O), 3 (C₄H₆O) have been prepared by the routs shown below. Suggest structures for these isomers (15 points)



1: ¹H NMR: 5.35(2H,s) , 1.0(4H,s); ¹³C NMR: 3different signals

2: IR: no strong peaks outside fingerprint; ¹H NMR: 3.0(2H, s) , 0.85(4H, t)

3: IR: 1770 cm⁻¹ , ¹H NMR: 3.02(4H, t, J=5Hz), 1.98(2H, m, J=5Hz)

6. Propose a mechanism for the following reaction(10 points):

